

DIRECTIONS for questions 1 to 5: Answer the questions on the basis of the information given below.

Seven persons – A through G – each of a different height, are standing in a line, from left to right, all facing the same direction. They are standing such that there are exactly three persons for each of whom, there are two persons standing adjacent to him, with both of them being taller than he is.

It is also known that

- i. A, who is not the shortest, is standing adjacent to the third tallest person in the group.
- ii. B, who is the fourth tallest person, is standing adjacent to the tallest person, but is not standing adjacent to the second tallest person.
- iii. D, who is not standing at any end, is standing four places to the left of the second tallest person.
- iv. F is standing to the immediate left of B.
- v. there are at least three persons who are both taller than C and shorter than E.

Q1. DIRECTIONS for questions 1 to 5: Select the correct alternative from the given choices.

Who is the sixth tallest person?

- ☐ a) **D**
- ☐ b) **C**
- ☐ c) **A**
- ☐ d) **G**

Q2. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.
Who among the following is standing two places away from the third tallest person?

- ☐ a) **B**
- ☐ b) **C**
- ☐ c) **G**
- ☐ d) **E**

Q3. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.

If the seven persons are ranked from 1 to 7 in the ascending order of their heights, how many pairs of persons standing adjacent to each other have consecutive ranks?

☐ a) 0

☐ b) 1

☐ c) 3

☐ d) 2

Q4. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.

If the seven persons are ranked from 1 to 7 in the ascending order of their heights, what is the maximum difference between the ranks of any two persons standing adjacent to each other?

☐ a) 5

☐ b) 3

☐ c) 4

☐ d) 6

Q5. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.

How many persons standing to the left of G are taller than him?

☐ a) 1

☐ b) 3

☐ c) 2

☐ d) 4

DIRECTIONS for questions 6 to 9: Answer the questions on the basis of the information given below.

During a particular month, Ramu, a food critic, visited exactly seven restaurants, A through G, and ranked the seven restaurants from 1 to 7 such that a numerically lower rank is considered better than a numerically higher rank. He visited at most one restaurant on any day of the month. He visited the restaurants ranked second, third and sixth on Mondays; he visited the restaurants ranked first and fifth on Fridays; he visited the restaurants ranked fourth and seventh on Saturdays.

It is also known that

- i. C, which he did not visit on a Saturday, was ranked worse than B.
- ii. the day of the week on which he visited F was not the same as the day of the week on which he visited D.
- iii. D was ranked worse than A and he did not visit D on a Monday.
- iv. three restaurants were ranked better than E but worse than G, but he did not visit these two restaurants on the same day.
- v. he did not visit A on a Monday and A was ranked worse than F.

Q6. DIRECTIONS for questions 6 to 9: Select the correct alternative from the given choices.

How many restaurants were ranked better than C?

- ☐ a) 1
- ☐ b) 2
- ☐ c) 4
- ☒ d) 5 ✓ Your answer is correct

Q7. DIRECTIONS *for questions 6 to 9:* Select the correct alternative from the given choices.
Which of the following restaurants is ranked better than G?

- ☐ a) **A**
- ☐ b) **B**
- ☒ c) **F** **✗ Your answer is incorrect**
- ☐ d) **More than one of the above**

Q8. DIRECTIONS *for questions 6 to 9:* Select the correct alternative from the given choices.

The ranks of any two restaurants that Ramu visited consecutively differed by more than one. Also, he visited the first restaurant on a Monday and visited a restaurant on every Friday, Saturday and Monday that followed (until he visited all the seven restaurants).

Which of the following can be the number of restaurants that Ramu visited before visiting A?

- ☐ a) **2**
- ☐ b) **3**
- ☐ c) **5**
- ☐ d) **More than one of the above**

Q9. DIRECTIONS *for questions 6 to 9:* Select the correct alternative from the given choices.

The ranks of any two restaurants that Ramu visited consecutively differed by more than one. Also, he visited the first restaurant on a Monday and visited a restaurant on every Friday, Saturday and Monday that followed (until he visited all the seven restaurants).

What is the difference between the ranks of the restaurant that he visited first and A?

- ☐ a) 1
- ☐ b) 2
- ☐ c) 3
- ☐ d) **Cannot be determined**

DIRECTIONS *for questions 10 to 14:* Answer the questions on the basis of the information given below.

A shoe manufacturing company hired exactly 300 employees at the beginning of June 2018. At the time of their hiring, each employee was skilled in exactly one activity among Polishing, Buffing and Stitching.

After they were hired, during each month, starting from June 2018 until March 2019, exactly 20 employees were taught Polishing, exactly 10 employees were taught Buffing and exactly 30 employees were taught Stitching.

It is known that, until the end of October 2018, no employee was skilled in more than two activities, while by the end of March 2019, all the employees were skilled in all the three activities.

Q10. DIRECTIONS *for questions 10 and 11:* Type in your answer in the input box provided below the question.

At the end of October 2018, how many employees were skilled in Polishing but not in Buffing?

Q11. DIRECTIONS *for questions 10 and 11:* Type in your answer in the input box provided below the question.

At the end of December 2018, how many employees were not skilled in Stitching but were skilled in Buffing?

Q12. DIRECTIONS *for questions 12 to 14:* Select the correct alternative from the given choices.

In any month, say 'xyz', if a_{xyz} represents the number of employees skilled in Polishing who were taught Stitching in that month and b_{xyz} represents the number of employees skilled in Buffing who were taught Stitching in that month, what is the maximum number of months, from June 2018 to October 2018, for which a_{xyz} is equal to b_{xyz} ?

☐ a) 5

☐ b) 4

☐ c) 2

☐ d) 3

Q13. DIRECTIONS *for questions 12 to 14:* Select the correct alternative from the given choices.

In any month say 'xyz', if a_{xyz} represents the number of employees skilled in only Polishing at the end of that month and b_{xyz} represents the number of employees skilled in only Buffing at the end of that month, what is the maximum number of months for which a_{xyz} is greater than b_{xyz} ?

- ☐ a) 0
- ☐ b) 1
- ☐ c) 2
- ☐ d) More than 2

Q14. DIRECTIONS *for questions 12 to 14:* Select the correct alternative from the given choices.

In any month say 'xyz', if a_{xyz} represents the number of employees skilled in Polishing who were taught Stitching, what is the maximum number of months, from June 2018 to March 2019, for which $a_{xyz} = 0$?

- ☐ a) 3
- ☐ b) 2
- ☐ c) 5
- ☐ d) None of the above

DIRECTIONS for questions 15 to 18: Answer the questions on the basis of the information given below.

Raghu, a Mathematics teacher, devised an assignment for the six children, A through F, in his class.

He gave two distinct numbers (both greater than 1) to each child and asked each child to multiply the two numbers that he was given. There were exactly six distinct numbers among the numbers that he gave to the six children and each number was given to not more than two children.

The following table provides the products obtained by the six children:

Child	Product
A	135
B	270
C	90
D	63
E	50
F	70

Q15. DIRECTIONS for question 15: Select the correct alternative from the given choices.

Which of the following numbers was given to both A and C?

- ☐ a) 5
- ☐ b) 9
- ☐ c) 15
- ☐ d) **None of the above**

Q16. DIRECTIONS *for questions 16 and 17: Type in your answer in the input box provided below the question.*

What is the sum of the twelve numbers given to all the children?

Q17. DIRECTIONS *for questions 16 and 17: Type in your answer in the input box provided below the question.*

For how many children is the sum of the numbers given to them greater than 20?

Q18. DIRECTIONS *for question 18:* Select the correct alternative from the given choices.
For which child is the difference between the two numbers given to him the highest?

☐ a) **B**

☐ b) **C**

☐ c) **E**

☐ d) **A**

DIRECTIONS for questions 19 to 22: Answer the questions on the basis of the information given below.

In a school, there were a certain number of students in each year from 2015 to 2019. Of the students in the school in any year, a certain number of the students were boys, while the rest were girls. The total number of students in the school in each year was distinct. Further, in any year, the number of boys in the school was not the same as the number of girls in the school.

Also, for each year from 2016 to 2019, the number of boys in the school in that year was the same as the number of girls in the school in the previous year.

It is also known that

- i. the number of girls in the school in 2015 was 30 less than the number of students in the school in 2017, while the number of boys in the school in 2015 was three-fifths the number of students in the school in 2019.
- ii. the number of students in the school in one of the years was 320, while in 2019, the number of boys in the school was 160.
- iii. the number of boys in the school in 2018 was 130, while the number of girls in the school in 2015 was 140.

Q19. DIRECTIONS for questions 19 and 20: Select the correct alternative from the given choices.

What is the difference between the number of boys in the school in the year 2015 and the number of girls in the school in the year 2016?

- ☐ a) 100
- ☐ b) 80
- ☐ c) 120
- ☐ d) 140

Q20. DIRECTIONS *for questions 19 and 20:* Select the correct alternative from the given choices.

What is the total number of students in the school in 2017?

☐ a) 200

☐ b) 170

☐ c) 190

☐ d) 150

Q21. DIRECTIONS *for questions 21 and 22:* Type in your answer in the input box provided below the question.

What is the minimum number of students who must have enrolled in the school in the year 2018?

Q22. DIRECTIONS *for questions 21 and 22:* Type in your answer in the input box provided below the question.

What is the minimum number of students who must have enrolled in the school in the year 2019?